DA roadmap

Data Science & AI (source: CourseDetails_DA.pdf, 74 entries).

How to use a course file

1. Watch the 🎓 **Main course** playlist in order, following the **Study plan** table (it links each week to the exact lecture videos).
2. If a lecturer doesn't work for you, switch to one of the 🎥 **Other full lecture series**; use ⚡ **Short videos** for intuition before or after a hard topic.
3. Read the book chapters and do the **Practice** project. Watching alone is not learning.

About the links: every YouTube playlist was opened on 2026-10-22; its title, video count and lecture list are copied from YouTube, not typed by hand. Official channels (NPTEL, IIT Madras, MIT OCW, Stanford Online, CMU, Harvard and others) were preferred; uploads by other people are marked *re-upload*. In the Books section, ✅ means the link was checked in a web search; other book/website links were not re-checked, so confirm they are still online.

Tier A = core, do it. Tier B = recommended elective. Tier C = advanced, only after its prerequisites.

Order of courses

Phase 1

Course	Tier	Needs first	Effort	Main video course	File
DA1300+DA1301 Programming and Data Structures (+ Python lab)	A	Nothing	10-12 weeks, ~8 h/week	Programming in Python	courses/P1_DA1300_DA1301_programming-and-data-structures-python-lab.pdf
DA1001 Applied Linear Algebra	A	School maths	12 weeks, ~6 h/week	Applied Linear Algebra	courses/P1_DA1001_applied-linear-algebra.pdf
DA1000 Calculus for Engineers	A	School maths	10-12 weeks	MIT 18.01 Single Variable Calculus, Fall 2006	courses/P1_DA1000_calculus-for-engineers.pdf

Phase 2

Course	Tier	Needs first	Effort	Main video course	File
DA1004 Probability and Statistics for Engineers	A	DA1000	12 weeks, ~7 h/week	Statistics 110: Probability	courses/P2_DA1004_probability-and-statistics-for-engineers.pdf
DA1002+DA1003 Optimization for Engineers (+ lab)	A	DA1000, DA1001	12 weeks	Stanford EE364A Convex Optimization I Stephen Boyd I 2023	courses/P2_DA1002_DA1003_optimization-for-engineers-lab.pdf
DA2303 Algorithms for Data Science	A	DA1300, DA1004	12-14 weeks	Programming, Data Structures and Algorithms using Python_Prof. Madhavan Mukund	courses/P2_DA2303_algorithms-for-data-science.pdf

Phase 3

Course	Tier	Needs first	Effort	Main video course	File
DA2304 Introduction to Computer Systems	A	DA1300	12 weeks	Foundations to Computer Systems Design	courses/P3_DA2304_introduction-to-computer-systems.pdf
DA2400+DA2401 Machine Learning I (+ lab)	A	DA1001, DA1004, DA1300, DA1002	14 weeks, ~10 h/week	Machine Learning Foundations	courses/P3_DA2400_DA2401_machine-learning-i-lab.pdf
DA2402 Data Curation and Visualization	A	DA1301	8 weeks	Data Science for Engineers / IIT Madras	courses/P3_DA2402_data-curation-and-visualization.pdf

Phase 4

Course	Tier	Needs first	Effort	Main video course	File
DA2403+DA2404 Artificial Intelligence (+ lab)	A	DA2400, DA2303	12 weeks	Artificial Intelligence Search Methods For Problem Solving	courses/P4_DA2403_DA2404_artificial-intelligence-lab.pdf
DA3409+DA3410 Deep Learning (+ lab)	A	DA2400, DA1000, DA1001	12-14 weeks	Deep Learning	courses/P4_DA3409_DA3410_deep-learning-lab.pdf
DA3406 Data Management	A	DA2303, DA2402	10-12 weeks	Database Management Systems	courses/P4_DA3406_data-management.pdf

Phase 5

Course	Tier	Needs first	Effort	Main video course	File
DA3407 Machine Learning II	A	DA2400, DA3409	12 weeks	CORNELL CS4780 "Machine Learning for Intelligent Systems"	courses/P5_DA3407_machine-learning-ii.pdf
DA3451+DA3452 Computer Vision / Speech & Language Technology (Core Basket I)	B	DA3409	12 weeks each	Stanford CS231N Deep Learning for Computer Vision I 2025 Stanford CS224N Natural Language Processing with Deep Learning I Spring 2024 I Professor Christopher Manning	courses/P5_DA3451_DA3452_computer-vision-speech-language-technology-core-.pdf
DA3408+DA5402 AI Operations / MLOps Labs	A	DA3406, DA3409	10 weeks	Machine Learning Operations (MLOps) - Rangarajan Vasudevan	courses/P5_DA3408_DA5402_ai-operations-mlops-labs.pdf
DA5500W Data Mining: Concepts and Techniques	B	DA2402, DA2400	10 weeks	Data Mining for Decision Making	courses/P5_DA5500W_data-mining-concepts-and-techniques.pdf

Phase 6

Course	Tier	Needs first	Effort	Main video course	File
DA6360 Advanced Optimisation for Machine Learning	C	DA1002, DA3409	12 weeks	Stanford EE364A Convex Optimization I Stephen Boyd I 2023	courses/P6_DA6360_advanced-optimisation-for-machine-learning.pdf
DA6800 Generative AI	B	DA3409	8-10 weeks	Stanford CS236: Deep Generative Models I 2023 I Stefano Ermon	courses/P6_DA6800_generative-ai.pdf

Phase 7

Course	Tier	Needs first	Effort	Main video course	File
DA6300 Quantum Computing and Machine Learning	C	DA1001	12 weeks	Introduction to Quantum Computing and Quantum Hardware	courses/P7_DA6300_quantum-computing-and-machine-learning.pdf
DA6336+DA7450 Privacy in AI / AI Privacy	C	DA1004, DA3409	10 weeks	A Course In Differential Privacy	courses/P7_DA6336_DA7450_privacy-in-ai-ai-privacy.pdf
DA5453 Learning from Human Preferences	C	DA1004, DA6800	10 weeks	The Post-Training Course (w/ Nathan Lambert's RLHF Book)	courses/P7_DA5453_learning-from-human-preferences.pdf

Not doable (or not worth doing) as self-study

Course(s)	Why
DA1200 Basics of Engineering Principles	Engineering-calculation/units course; low value as self-study.
DA1100, DA2101, DA2102 (comp. chemistry/physics/biology)	Doable but off-track; pick one only if you like the subject.
RCDA*, RC9008A, DA5412, DA6900-6902, DA4902, DA4912, DA5410	Projects, research credits and industrial placement.
DA5111-5113, DA5210-5223 (Manufacturing analytics)	Industry-domain course set with company/case material.
DA6701 (Finance), DA6050 (Public AI), DA5452 (Estimation and control), DA6100 (AI systems for science)	Domain-specific; revisit if it matches an interest.

Course(s)	Why
DA6999, DA7999, DA7400, DA7016	Research seminars on recent papers.
Responsible AI, Online & RL, Core Basket II, Project I/II (sem-7 DA7xxx)	No course sheet in the PDF.